

1. A previously healthy 6-year-old boy is brought to the office because of a 2-day history of right ear pain. He went swimming in a lake 3 days ago. On examination, the right ear canal appears normal with no pain on motion of the canal or tragus; the eardrum is red, bulging, and immobile. The left ear and pharynx appear normal. The neck is supple; there is no lymphadenopathy. The remainder of the examination shows no abnormalities. Which of the following is the most likely diagnosis?
- ☐ A) Acute otitis media
 - ☐ B) Bullous otitis
 - ☐ C) Foreign body
 - ☐ D) Labyrinthitis
 - ☐ E) Otitis externa
 - ☐ F) Otitis media with effusion



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2. A 22-month-old girl with recurrent cough and shortness of breath is brought to the physician for a follow-up examination. She was born at term and weighed 3200 g (7 lb 1 oz). Pregnancy was uncomplicated. Recurrent episodes of wheezing with cough began after the age of 6 months and were successfully treated with combinations of oral antibiotics and β_2 -adrenergic agonists. She was admitted to the hospital for treatment of lobar pneumonia at the ages of 9, 12, and 16 months. She was treated for rectal prolapse 2 months ago. There is a family history of childhood-onset asthma of varying severity persisting into adulthood. She is alert and acyanotic. She weighs 10.4 kg (23 lb). Her temperature is 37.3°C (99.2°F), pulse is 115/min, and respirations are 52/min. Examination shows mild subcostal and intercostal retractions. Diffuse inspiratory crackles are heard with hyperresonance to percussion. There is minimal clubbing of the nail beds. An x-ray of the chest shows generalized hyperinflation, bronchial thickening in the upper lobes, and patchy atelectasis throughout the lungs. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) Measurement of serum IgE concentration
 - ☐ B) PPD skin test
 - ☐ C) Sweat chloride test
 - ☐ D) Methacholine bronchial challenge test
 - ☐ E) Nasal mucosal biopsy

3. A 2495-g (5-lb 8-oz) female newborn has tachypnea and intercostal retractions 25 minutes after delivery at 32 weeks' gestation to a 23-year-old primigravid woman. Membranes ruptured 1 hour prior to delivery; no meconium was noted. Apgar scores were 7 and 7 at 1 and 5 minutes, respectively. The mother is afebrile. Recent vaginal culture of the mother was negative for group B streptococcus. The patient appears pink. Her temperature is 36.8°C (98.2°F), pulse is 150/min, respirations are 70/min, and blood pressure is 60/30 mm Hg. Examination shows perioral cyanosis, nasal flaring, grunting, and intercostal retractions. Breath sounds are decreased bilaterally. Pulse oximetry on room air shows an oxygen saturation of 84%. After administration of oxygen (4 L/min) by face mask, oxygen saturation increases to 91%. An x-ray of the chest shows diffuse, bilateral haziness in the lungs. Which of the following is the most likely diagnosis?
- ☐ A) Bacterial pneumonia
 - ☐ B) Congenital emphysema
 - ☐ C) Cyanotic congenital heart disease
 - ☐ D) Hyaline membrane disease
 - ☐ E) Meconium aspiration
 - ☐ F) Transient tachypnea of the newborn



4. A previously healthy 6-month-old boy is brought to the emergency department because of a 1-day history of progressively decreasing activity level and oral intake. His mother reports that he has had fewer wet diapers today. He has not had vomiting or diarrhea. His temperature is 37°C (98.6°F), pulse is 245/min and regular, respirations are 32/min and shallow, and blood pressure is 60/40 mm Hg. He appears gray and poorly perfused; pulse oximetry is not obtainable. He is lethargic but responsive to pain. Lungs are clear to auscultation. Abdominal examination shows no abnormalities. Administration of supplemental oxygen is begun. He is placed on a cardiorespiratory monitor. An ECG rhythm strip is shown. Which of the following is the most appropriate next step in management?
- ☐ A) Intramuscular injection of ceftriaxone
 - ☐ B) Administration of a bolus of 0.9% saline
 - ☐ C) Chest compressions
 - ☐ D) Cardioversion
 - ☐ E) Endotracheal intubation

5. A 12-year-old girl is brought to the physician because of vaginal bleeding for 2 weeks. Pubertal development began at the age of 10 years. Her first menstrual period began 6 weeks ago and lasted for 10 days; she used 10 to 12 sanitary pads daily. Her second menstrual period began 2 weeks ago and has not stopped. She had been using 12 pads daily until 2 days ago; since then, she has used 6 daily. Her temperature is 37°C (98.6°F), pulse is 100/min, respirations are 16/min, and blood pressure is 95/60 mm Hg. Examination shows no other abnormalities. Laboratory studies show:

Hemoglobin	10 g/dL
Leukocyte count	10,000/mm ³ with a normal differential
Platelet count	180,000/mm ³
Prothrombin time	12 sec (INR=1)
Partial thromboplastin time	45 sec

Which of the following is the most likely cause of these findings?

- ☐ A) Acute lymphocytic leukemia
- ☐ B) Anovulatory menstrual cycles
- ☐ C) Factor XI deficiency
- ☐ D) Factor XIII deficiency
- ☐ E) Hyperthyroidism
- ☐ F) von Willebrand disease

6. A 2-week-old newborn is brought for a well-child examination. He was born at 38 weeks' gestation to a 27-year-old primigravid woman following an uncomplicated pregnancy and delivery; he weighed 2296 g (5 lb 1 oz). He is at the 5th percentile for length, weight, and head circumference. Examination shows short palpebral fissures, maxillary hypoplasia, micrognathia, and a thin upper lip. The lungs are clear to auscultation. A grade 1/6 systolic murmur is heard at the left sternal border with radiation to the left axilla and back. Femoral pulses are normal. Neurologic examination shows mild hypotonia. Which of the following is the most likely diagnosis?
- ☐ A) Congenital toxoplasmosis
 - ☐ B) Down syndrome
 - ☐ C) Fetal alcohol syndrome
 - ☐ D) Fragile X syndrome
 - ☐ E) Seminiferous tubule dysgenesis (Klinefelter syndrome)
 - ☐ F) Trisomy 13 syndrome

7. A 15-month-old girl is brought to the physician because she has not begun to crawl or walk. She has been exclusively breast-fed since birth. She is at the 50th percentile for length and the 85th percentile for weight. Her language and social skills are appropriate for age. Examination shows no abnormalities except for widening of the wrists. Which of the following serum findings is most likely to be increased?
- ☐ A) ALT activity
 - ☐ B) Alkaline phosphatase activity
 - ☐ C) Calcium concentration
 - ☐ D) Magnesium concentration
 - ☐ E) Phosphorus concentration

8. A 6-year-old boy is brought to the clinic by his mother because of intermittent bed-wetting. He was toilet trained at the age of 4 years but has never remained dry through the night and typically wets the bed three to four times weekly. He does not wet himself during the day. He has not had pain with urination or urinary urgency, hesitancy, or dribbling. Urine stream is normal. His mother says that he does not snore, but that he is a “heavy sleeper.” He has not had fever, diarrhea, or constipation. He recently started first grade and has not yet made any friends. Growth and development are otherwise appropriate for age. Vital signs are within normal limits. The abdomen is soft and nontender. The penis is circumcised. The testes are descended. Urinalysis shows:

pH	7.0
Specific gravity	1.020
Blood	negative
Glucose	negative
Protein	negative
Ketones	negative
Nitrites	negative
Leukocyte esterase	negative

Which of the following is the most likely explanation for this patient's enuresis?

- ☐ A) Adjustment disorder
- ☐ B) Detrusor muscle overactivity
- ☐ C) Maturational delay
- ☐ D) Primary sleep disorder
- ☐ E) Urinary tract infection

9. An 18-year-old woman comes to the physician because she has never had a menstrual period. She takes no medications. She is 163 cm (5 ft 4 in) tall and weighs 60 kg (132 lb); BMI is 23 kg/m². Her temperature is 37°C (98.6°F), pulse is 80/min, respirations are 20/min, and blood pressure is 110/60 mm Hg. Breast development is sexual maturity rating stage 1, and pubic hair development is sexual maturity rating stage 3. Pelvic examination shows a normal vagina and a small uterus. Her serum follicle-stimulating hormone concentration is 92 mIU/mL. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) Karyotype analysis
 - ☐ B) Measurement of serum progesterone concentration
 - ☐ C) Vaginal cytology
 - ☐ D) Hysterosalpingography
 - ☐ E) CT scan of the head

10. A 3-month-old female infant born at 34 weeks' gestation is admitted to the hospital in January because of a 3-day history of copious rhinorrhea, fever, poor feeding, and periods of tachypnea. Her 3-year-old brother was diagnosed with an upper respiratory tract infection 5 days ago. Her temperature is 38°C (100.4°F), respirations are 65/min, and blood pressure is 60/40 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 92%. Physical examination shows intercostal retractions. Wheezes are heard bilaterally. A chest x-ray shows hyperinflation and peribronchial thickening. Which of the following is the most likely pathogen?
- ☐ A) *Bordetella pertussis*
 - ☐ B) *Chlamydia trachomatis*
 - ☐ C) Influenza virus
 - ☐ D) *Listeria monocytogenes*
 - ☐ E) *Mycoplasma pneumoniae*
 - ☐ F) Respiratory syncytial virus
 - ☐ G) *Staphylococcus aureus*
 - ☐ H) *Streptococcus pneumoniae*



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11. A previously healthy 4-year-old boy is brought to the office because of a 3-day history of easy bruising that began on the extremities but now includes the trunk. There is no history of trauma. His parents say that he is very active. His temperature is 37°C (98.6°F), pulse is 100/min, and respirations are 16/min. Examination shows scattered petechiae and multiple ecchymoses in various stages of healing over the trunk and extremities; they are most pronounced on the extremities. Laboratory studies show:

Hemoglobin	12 g/dL
Leukocyte count	6000/mm ³
Platelet count	1000/mm ³
Prothrombin time	12 sec (control 12.2) (INR=1)
Partial thromboplastin time	34 sec (control 33)

Which of the following is the most likely cause of this patient's condition?

- ☐ A) Abnormal platelet function
- ☐ B) Child abuse
- ☐ C) Decreased platelet production
- ☐ D) Disseminated intravascular coagulation
- ☐ E) Increased platelet destruction



12. A 10-year-old girl is brought to the physician because of a 6-month history of increased facial hair growth and facial acne. She is at the 50th percentile for height and weight. Her temperature is 37°C (98.6°F), pulse is 64/min, respirations are 12/min, and blood pressure is 110/60 mm Hg. Examination shows mild clitoral enlargement and an early male pubic hair pattern. Laboratory studies show a serum testosterone concentration of 0.2 nmol/L (N=0.17–0.7), dehydroepiandrosterone sulfate concentration of 9 µg/mL (N=0.5–5.4), and urine 17-ketosteroid excretion of 20 mg/24 h (N=6–15). Pelvic ultrasonography shows no abnormalities. Which of the following is the most likely diagnosis?
- ☐ A) Adrenal adenoma
 - ☐ B) Androgen insensitivity syndrome
 - ☐ C) Ovarian adenoma
 - ☐ D) Pituitary adenoma
 - ☐ E) Polycystic ovarian syndrome



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13. A 2-day-old male newborn is evaluated in the hospital for hydronephrosis first seen on prenatal ultrasonography at 28 weeks' gestation. He was born at term following spontaneous vaginal delivery. He feeds well from a bottle. Vital signs are within normal limits. He voids during the examination; urine output is normal, but urine flow is poor with dribbling of urine from the penis. Postnatal ultrasonography shows bilateral hydronephrosis and hydroureter and a thick-walled bladder. Which of the following is the most likely diagnosis?
- ☐ A) Duplicated renal collecting system
 - ☐ B) Neurogenic bladder
 - ☐ C) Posterior urethral valve
 - ☐ D) Ureteropelvic junction obstruction
 - ☐ E) Vesicoureteral reflux

14. A previously healthy 15-month-old boy is brought to the physician in late winter by his mother because of persistent vomiting followed by fever and diarrhea. His symptoms began 2 days ago after a large family gathering. His mother tried to give him apple juice, but he vomited the juice after drinking only 4 ounces. The patient is crying and appears alert. His temperature is 39°C (102.2°F), pulse is 120/min, respirations are 36/min, and blood pressure is 88/56 mm Hg. Capillary refill time is normal. Examination shows tear production and slightly dry mucous membranes. The abdomen is nontender; bowel sounds are hyperactive. Test of the stool for occult blood is negative. Which of the following is the most appropriate treatment for this patient?
- ☐ A) Oral bismuth subsalicylate
 - ☐ B) Oral hydration solution
 - ☐ C) Oral loperamide
 - ☐ D) Rectal prochlorperazine
 - ☐ E) Intravenous 5% dextrose in water

15. A 4-month-old girl is brought to the physician for a well-child examination in October. Pregnancy and delivery were uncomplicated. Immunizations are up-to-date. Growth and development are appropriate for age. Vital signs are within normal limits. Examination shows no abnormalities. Immunization against which of the following is most appropriate at this time?

- ☐ A) Chickenpox
- ☐ B) Influenza
- ☐ C) Measles-mumps-rubella
- ☐ D) Rotavirus
- ☐ E) Shingles



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16. A 15-year-old boy is brought to the emergency department 1 hour after the onset of nausea and vomiting, weakness, and headache that began during practice for a cross-country meet. On arrival, he is mildly disoriented. One week ago, he had viral gastroenteritis and missed 4 days of practice. He returned to practice 3 days ago. Today, the outside temperature has been near 90°F. His coach reports that he observed him drinking large amounts of water before and during practice. The patient reports that he skipped lunch today. He is at the 25th percentile for height and weight. His temperature is 37.6°C (99.7°F), pulse is 100/min, and blood pressure is 110/75 mm Hg. Examination shows edema of the hands and feet. Which of the following is the most likely explanation for this patient's symptoms?
- ☐ A) Dehydration
 - ☐ B) Heat exhaustion
 - ☐ C) Heat stroke
 - ☐ D) Hypoglycemia
 - ☐ E) Hyponatremia



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17. An 18-month-old boy is brought to the physician because of poor linear growth. There is no history of vomiting or diarrhea. His blood pressure is 90/60 mm Hg. Laboratory studies show:

Serum	
Na ⁺	138 mEq/L
Cl ⁻	112 mEq/L
K ⁺	3.1 mEq/L
HCO ₃ ⁻	16 mEq/L
Urine pH	7.6

Arterial blood gas analysis on room air shows:

pH	7.28
Pco ₂	26 mm Hg
Po ₂	98 mm Hg

Which of the following is the most likely cause of this patient's hypokalemia?

- ☐ A) Addison disease
- ☐ B) Congenital adrenal hyperplasia
- ☐ C) Hyperparathyroidism
- ☐ D) Juxtaglomerular cell hyperplasia
- ☐ E) Pheochromocytoma



K ⁺	3.1 mEq/L
HCO ₃ ⁻	16 mEq/L
Urine pH	7.6

Arterial blood gas analysis on room air shows:

pH	7.28
Pco ₂	26 mm Hg
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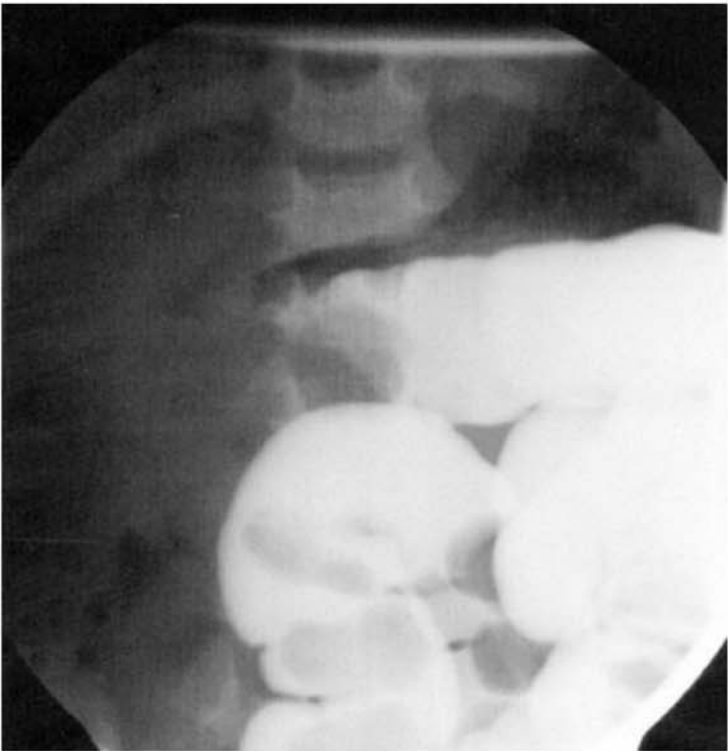
Which of the following is the most likely cause of this patient's hypokalemia?

- ☐ A) Addison disease
- ☐ B) Congenital adrenal hyperplasia
- ☐ C) Hyperparathyroidism
- ☐ D) Juxtaglomerular cell hyperplasia
- ☐ E) Pheochromocytoma
- ☐ F) Primary hyperaldosteronism
- ☐ G) Pseudohypoaldosteronism
- ☐ H) Renal artery stenosis
- ☐ I) Renal tubular acidosis



18. A 3-day-old girl undergoes evaluation prior to discharge from the hospital. She was born at 39 weeks' gestation following an uncomplicated pregnancy and cesarean delivery for cephalopelvic disproportion and weighed 3256 g (7 lb 3 oz). Examination at birth showed no abnormalities. She is exclusively breast-fed. She is alert and active. Her temperature is 37.1°C (98.8°F), pulse is 104/min, and respirations are 36/min. Pulse oximetry on room air shows an oxygen saturation of 99%. A new grade 2/6, early systolic murmur is heard in the pulmonic area with radiation to the lung fields and is loudest in the right axilla. Peripheral pulses are normal. Which of the following is the most likely diagnosis?
- ☐ A) Aortic stenosis
 - ☐ B) Atrial septal defect
 - ☐ C) Patent ductus arteriosus
 - ☐ D) Pulmonary stenosis
 - ☐ E) Ventricular septal defect





19. A previously healthy 10-month-old boy is brought to the emergency department by his mother because of a 10-hour history of bloody, mucoid stools. He also has vomited three times during this time. His mother says that he cries and draws up his legs during the episodes of vomiting. Between episodes, he appears comfortable. He is playful and is not in distress. His temperature is 37.1°C (98.8°F), pulse is 110/min, respirations are 24/min, and blood pressure is 96/58 mm Hg. Abdominal examination shows mild distention. A 4-cm, tender, elongated mass is palpated in the right upper quadrant. There is no



19. A previously healthy 10-month-old boy is brought to the emergency department by his mother because of a 10-hour history of bloody, mucoid stools. He also has vomited three times during this time. His mother says that he cries and draws up his legs during the episodes of vomiting. Between episodes, he appears comfortable. He is playful and is not in distress. His temperature is 37.1°C (98.8°F), pulse is 110/min, respirations are 24/min, and blood pressure is 96/58 mm Hg. Abdominal examination shows mild distention. A 4-cm, tender, elongated mass is palpated in the right upper quadrant. There is no hepatosplenomegaly. Rectal examination shows bloody mucoid stool in the rectal vault. Results of a contrast enema are shown. Which of the following is the most likely explanation for these findings?
- ☐ A) Bleeding diverticulum
 - ☐ B) Intussusception
 - ☐ C) Malrotation volvulus
 - ☐ D) Peptic ulcer
 - ☐ E) Pyloric stenosis

20. A previously healthy 13-year-old boy is brought to the physician because of a 4-week history of intermittent left groin pain associated with a limp. He has not had fever. He is not in acute distress. He is at the 50th percentile for height, 97th percentile for weight, and above the 97th percentile for BMI. His temperature is 37°C (98.6°F), pulse is 90/min, respirations are 20/min, and blood pressure is 116/75 mm Hg. On examination, internal rotation and abduction of the left hip are limited. Internal and external rotation to 75 degrees produces pain. The left hip externally rotates on passive flexion. Examination of the right hip and both knees shows no abnormalities. AP and cross-table lateral x-rays of the left hip show irregular widening of the epiphyseal plate of the left femur and mild osteopenia of the left femoral neck and head. In addition to analgesic therapy and discussion about weight loss, which of the following is the most appropriate next step in management?

- ☐ A) Physical therapy
- ☐ B) Abduction bracing
- ☐ C) Traction
- ☐ D) Spica cast immobilization
- ☐ E) Internal fixation

21. A 9-month-old boy is brought to the office for a well-child examination. His mother says he is a happy baby. He likes to lie on the floor and can roll from his stomach to his back; he has been trying to roll from his back to his stomach over the past 2 weeks. He can sit with support. His appetite is good, and he likes to feed himself crackers and cereal from the high-chair table. He jabbbers, imitates sounds, and says "Dada" and "Mama" to everyone and everything. Immunizations are up-to-date. He is at the 75th percentile for length and 50th percentile for weight. Examination shows no abnormalities. Which of the following is the most accurate assessment of this patient's development?

	Fine Motor Development	Gross Motor Development	Language Development	Social Development
☐ A)	Delayed	delayed	delayed	delayed
☐ B)	Delayed	delayed	normal	normal
☐ C)	Delayed	normal	delayed	delayed
☐ D)	Delayed	normal	normal	normal
☐ E)	Normal	delayed	normal	normal
☐ F)	Normal	normal	delayed	normal
☐ G)	Normal	normal	normal	delayed
☐ H)	Normal	normal	normal	normal

22. A previously healthy 15-year-old girl is brought to the physician because of severe cramps during menses since menarche 6 months ago. Menses occur at regular 28-day intervals. Treatment with over-the-counter analgesics and nonsteroidal anti-inflammatory drugs has not relieved the pain. Examination, including pelvic examination, shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Reassurance
- ☐ B) Psychiatric assessment
- ☐ C) Narcotic analgesic therapy
- ☐ D) Oral contraceptive therapy
- ☐ E) Laparoscopy



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23. A previously healthy 10-year-old boy has a 10-day history of progressive cough, low-grade fever, and slight dyspnea on exertion. Examination shows diffuse crackles bilaterally. An x-ray of the chest shows a diffuse perihilar infiltrate. Which of the following is the most likely diagnosis?

- ☐ A) Pneumonia due to respiratory syncytial virus
- ☐ B) Pneumonia due to *Streptococcus pneumoniae*
- ☐ C) Pneumonia due to *Mycoplasma pneumoniae*
- ☐ D) Congestive heart failure
- ☐ E) Tuberculosis



24. A 3200-g (7-lb 1-oz) male newborn is delivered at 38 weeks' gestation. Pregnancy, labor, and delivery were uncomplicated. Apgar scores are 8 and 9 at 1 and 5 minutes, respectively. Examination shows no abnormalities except for bilateral flank masses. Laboratory studies show:

Hematocrit	50%
Serum	
Urea nitrogen	8 mg/dL
Creatinine	0.9 mg/dL
Urine	
pH	6
Specific gravity	1.009
Blood	moderate
Protein	trace

Ultrasonography shows bilateral enlargement of the kidneys with mixed echogenicity. Which of the following is the most likely diagnosis?

- ☐ A) Nephroblastoma (Wilms tumor)
- ☐ B) Neuroblastoma
- ☐ C) Polycystic kidney disease
- ☐ D) Renal cell carcinoma
- ☐ E) Renal vein thrombosis

25. A 10-year-old boy is brought to the office because of a 2-day history of redness and mild pain in his left eye. He is afebrile. Examination of the left eye shows a mildly erythematous conjunctiva. There is conjunctival edema and tenderness and a 3-mm nodule at the lower lateral lid margin. Examination of the right eye shows no abnormalities. Which of the following is the most appropriate treatment?

- ☐ A) Nasolacrimal massage
- ☐ B) Oral amoxicillin
- ☐ C) Topical prednisolone
- ☐ D) Topical tetrahydrozoline
- ☐ E) Warm compresses

26. A 16-year-old girl is brought to the emergency department because of a 3-day history of cough productive of thick yellowish sputum and a 1-day history of shortness of breath. She has not had fever. She has a history of asthma treated with albuterol and inhaled corticosteroids for which she required hospitalization on 10 occasions between the ages of 8 and 10 years. She has smoked one and one-half packs of cigarettes daily for 4 years. Her temperature is 37°C (98.6°F), pulse is 102/min, respirations are 42/min, and blood pressure is 108/80 mm Hg. Examination shows mild clubbing of the digits and increased anteroposterior chest diameter. Occasional end-expiratory wheezes are heard on auscultation. A chest x-ray shows no abnormalities other than severe bilateral hyperinflation. Which of the following is the most likely diagnosis?

- ☐ A) α_1 -Antitrypsin deficiency
- ☐ B) Congenital lobar emphysema
- ☐ C) Kartagener syndrome
- ☐ D) Pulmonary sequestration
- ☐ E) Sarcoidosis

27. A 5-year-old boy is brought to the office because of a 3-day history of a rash on his left leg. He has had five previous skin abscesses that have required prolonged treatment. He was admitted to the hospital for osteomyelitis at the ages of 2 and 4 years and pneumonia at the age of 3 years. His maternal uncle died of fungal pneumonia at the age of 5 years. The patient's temperature is 38.8°C (101.8°F). Examination shows several enlarged, tender left inguinal lymph nodes. There is a 2 × 3-cm warm, erythematous macular rash over the left lower extremity. Which of the following mechanisms of disease is the most likely cause of this patient's symptoms?
- ☐ A) Abnormal leukocyte adhesion
 - ☐ B) Abnormal leukocyte degranulation
 - ☐ C) Bone marrow infiltration
 - ☐ D) Impaired leukocyte chemotaxis
 - ☐ E) Impaired leukocyte microbicidal activity

28. A 12-year-old girl has had cramping abdominal pain, nausea, and four episodes of vomiting over the past 24 hours. Over the past 2 months, she has been drinking large quantities of fluids and has had urinary frequency; she has had a 6.8-kg (15-lb) weight loss during this period. She appears tired but is alert and answers questions appropriately. There is diffuse abdominal tenderness and a fruity odor to her breath. Which of the following is the most likely additional finding in this patient?
- ☐ A) Exophthalmos
 - ☐ B) Hypertension
 - ☐ C) Increased tidal volume
 - ☐ D) Neutropenia
 - ☐ E) Thrombocytopenia

29. A 2-month-old girl is brought to the physician by her mother for a well-child examination. Six weeks ago, she had watery diarrhea with blood in her stool. At that time, she was switched from a cow milk-based formula, which she had been fed since birth, to a soy-based formula, and the diarrhea resolved. She appears well. She is at the 50th percentile for length, weight, and head circumference. Examination shows no abnormalities. The mother should be counseled about which of the following regarding this patient's diet?
- ☐ A) She will likely have lifelong lactose intolerance
 - ☐ B) She will likely have lifelong milk protein intolerance
 - ☐ C) She will likely outgrow her lactose intolerance
 - ☐ D) She will likely outgrow her milk protein intolerance

30. A 10-year-old girl is brought to the physician because of a 5-day history of low-grade fever, mild generalized itching, and fatigue and a 1-day history of rash that started over her face and now involves her entire body. She also has had mild rhinorrhea. She has not had abdominal discomfort or diarrhea. She recently immigrated to the USA from Central America with her family. Her temperature is 37.8°C (100°F). Examination shows discrete, maculopapular lesions over the face, extremities, trunk, and back. There are no ocular or oral lesions. Which of the following vaccines is most likely to have prevented this patient's condition?

- ☐ A) Diphtheria-tetanus-acellular pertussis
- ☐ B) *Haemophilus influenzae* type b
- ☐ C) Hepatitis B
- ☐ D) Inactivated poliovirus
- ☐ E) Influenza virus
- ☐ F) Measles-mumps-rubella
- ☐ G) Pneumococcal
- ☐ H) Typhoid
- ☐ I) Varicella



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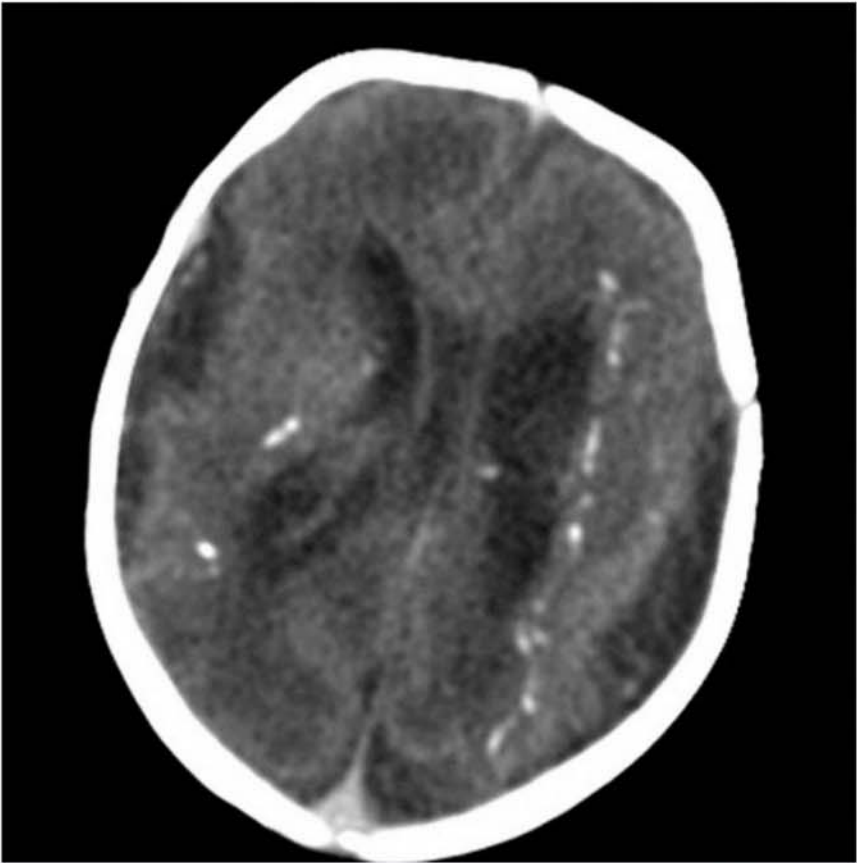


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31. A 7-year-old girl is brought to the physician by her mother because of a 2-day history of a burning sensation during urination. She has been treated with nitrofurantoin for urinary tract infections on two separate occasions; cultures at the time of the infections were negative. Examination shows a frothy vaginal discharge. The vulva is normal, and there is a small uterus. A wet mount preparation of the discharge shows *Trichomonas vaginalis*. Urinalysis shows no abnormalities except for occasional *T. vaginalis*. Which of the following is the most appropriate next step in management?
- ☐ A) Assessment for sexual abuse
 - ☐ B) Urine culture
 - ☐ C) Clotrimazole therapy
 - ☐ D) Intravenous pyelography
 - ☐ E) Cystoscopy





32. A female newborn is admitted to the neonatal intensive care unit because of diffuse jaundice and petechial rash noted at birth. She was born at 35 weeks' gestation following spontaneous vaginal delivery to a 23-year-old woman, gravida 1, para 1. Pregnancy was complicated by fetal growth restriction. There is no family history of serious illness. There are no pets in the home. The patient is below the 3rd percentile for length, weight, and head circumference. Vital signs are within normal limits. Examination shows diffuse jaundice and a petechial rash over the head, trunk, and all extremities. No murmurs are heard. The liver edge is palpated 5 cm below the right costal margin. The spleen tip is palpated 3 cm below the left costal margin. Laboratory studies show:

Hemoglobin	16 g/dL (N=15–24)
Leukocyte count	8300/mm ³ (N=9100–34,000)
Platelet count	78,000/mm ³
Serum	
Bilirubin, total	16 mg/dL
Direct	7.3 mg/dL

A CT scan of the head is shown. Brain stem auditory evoked potential testing shows sensorineural hearing loss bilaterally. Which of the following is most likely to confirm the diagnosis?

- ☐ A) Bone marrow aspiration and analysis
- ☐ B) Chromosomal microarray
- ☐ C) Lymphocyte enumeration by flow cytometry
- ☐ D) MRI of the brain
- ☐ E) Serum rubella IgM titer
- ☐ F) Serum *Toxoplasma gondii* IgM titer
- ☐ G) Urine viral culture

33. A 1-month-old male newborn is brought to the emergency department because of a 2-day history of forceful, nonbilious vomiting. His mother reports that the vomiting has been getting worse and that he now seems irritable and acts hungry afterward. He was born at term and weighed 2948 g (6 lb 8 oz). There were no prenatal or perinatal complications. He has not had any sick contacts and does not attend day care. His mother underwent pyloromyotomy as an infant. The patient's current weight is 2835 g (6 lb 4 oz). His temperature is 37°C (98.6°F), pulse is 180/min, respirations are 36/min, and blood pressure is 60/40 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 99%. On physical examination, the anterior fontanel is sunken and mucous membranes are dry. Which of the following sets of electrolyte findings is most likely in this patient?

	Na ⁺ (mEq/L)	K ⁺ (mEq/L)	Cl ⁻ (mEq/L)	HCO ₃ ⁻ (mEq/L)
☐ A)	136	4.2	84	24
☐ B)	136	4.7	84	38
☐ C)	140	3.9	100	24
☐ D)	145	4	105	38
☐ E)	145	5	110	24

34. A 17-year-old boy comes to the physician because of a 1-day history of swelling of his right hand. The swelling began 6 hours after he was involved in a fistfight at school the previous day. On questioning, he says he hit his opponent in the mouth and abdomen. Examination shows an erythematous, tender dorsum of the right hand. There is a small laceration over the fourth metacarpal head. Which of the following organisms are the most likely cause of this patient's symptoms?
- ☐ A) Aerobic gram-positive cocci
 - ☐ B) Anaerobic gram-positive cocci
 - ☐ C) Gram-negative bacilli
 - ☐ D) Gram-positive bacilli
 - ☐ E) Mixed flora

35. A 9-month-old boy is brought to the emergency department by his mother who reports that he injured his right thigh 2 hours ago when he slipped on a wet floor. Examination of the right thigh shows tenderness and swelling. Range of motion of the right hip is limited. An x-ray of the right lower extremity shows a spiral fracture of the femur. Which of the following is the most appropriate next step in management?

- ☐ A) Notification of child protective services
- ☐ B) Measurement of serum calcium concentration
- ☐ C) Complete blood count
- ☐ D) Bone scan of the right femur
- ☐ E) CT scan of the hip

36. A 2000-g (4-lb 7-oz) female newborn is jittery. She was born at 35 weeks' gestation to a 16-year-old primigravid patient. The membranes ruptured 6 hours before delivery. Her rectal temperature is 35.6°C (96°F). Examination shows no abnormalities except increased irritability. A heel-stick blood glucose concentration is 35 mg/dL. After administration of glucose water, a repeat heel-stick blood glucose concentration is 40 mg/dL. Blood is drawn for a complete blood count and culture. Urinalysis and cerebrospinal fluid analysis are ordered. Which of the following is the most appropriate pharmacotherapy for this newborn?
- ☐ A) Ampicillin and gentamicin
 - ☐ B) Ceftriaxone
 - ☐ C) Cilastatin-imipenem
 - ☐ D) Erythromycin
 - ☐ E) Penicillin G and tobramycin



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37. A 12-year-old girl is brought to the physician for a follow-up examination. She has a 3-month history of moderate asthma. Her pulse is 90/min, respirations are 20/min, and blood pressure is 110/70 mm Hg. Examination shows diffuse expiratory wheezes bilaterally. Which of the following is the most appropriate initial pharmacotherapy for this patient?
- ☐ A) Inhaled corticosteroid daily and inhaled short-acting β -adrenergic agonist as needed
 - ☐ B) Inhaled short-acting β -adrenergic agonist as needed
 - ☐ C) Intramuscular long-acting corticosteroid every 2 months
 - ☐ D) Oral corticosteroid daily
 - ☐ E) Oral corticosteroid and oral bronchodilator daily

38. A 10-year-old girl is brought to the physician for her first well-child examination. She was recently adopted from an orphanage in Eastern Europe. Her parents were informed that she has an unrepaired congenital heart defect, but further details are not available. Growth and development are appropriate for age. Her pulse is 96/min, respirations are 24/min, and blood pressure is 102/66 mm Hg in the right upper extremity and 114/72 mm Hg in the right lower extremity. Pulse oximetry on room air shows an oxygen saturation of 93%. Cardiac examination shows a hyperdynamic precordium with a right ventricular impulse. There is a loud, narrowly split S_2 . A grade 2/6 early to mid-systolic murmur is heard over the lower left sternal border. The liver edge is palpated 2 cm below the right costal margin. Brachial and femoral pulses are normal. An ECG shows right ventricular hypertrophy. The most likely mechanism of this patient's condition is an increase in which of the following?

- ☐ A) Central venous pressure
- ☐ B) Left ventricular end-diastolic pressure
- ☐ C) Pulmonary vascular resistance
- ☐ D) Systemic cardiac output
- ☐ E) Systemic vascular resistance



39. A previously healthy 2-year-old girl is brought to the physician because of a 2-day history of fatigue, red spots over her body, and facial swelling. During the past 24 hours, she has urinated only twice, although she has been drinking well. Seven days ago, she had fever and bloody diarrhea. She received supportive therapy, and her fever and diarrhea have resolved. Today, she appears pale. Her temperature is 37°C (98.6°F), pulse is 160/min, respirations are 28/min, and blood pressure is 120/80 mm Hg. Examination shows mild generalized edema. Laboratory studies show:

Hemoglobin	7 g/dL
Leukocyte count	12,000/mm ³
Platelet count	30,000/mm ³
Prothrombin time	11.5 sec (INR=1)
Partial thromboplastin time	32 sec
Serum	
Na ⁺	129 mEq/L
K ⁺	5.9 mEq/L
Cl ⁻	100 mEq/L
HCO ₃ ⁻	19 mEq/L
Urea nitrogen	30 mg/dL
Creatinine	3.2 mg/dL
Urine	
Blood	1+
Protein	none
RBC	none

Which of the following is the most likely explanation for this patient's oliguria?

☐ A) Decreased intravascular oncotic pressure



Leukocyte count	12,000/mm ³
Platelet count	30,000/mm ³
Prothrombin time	11.5 sec (INR=1)
Partial thromboplastin time	32 sec
Serum	
Na ⁺	129 mEq/L
K ⁺	5.9 mEq/L
Cl ⁻	100 mEq/L
HCO ₃ ⁻	19 mEq/L
Urea nitrogen	30 mg/dL
Creatinine	3.2 mg/dL
Urine	
Blood	1+
Protein	none
RBC	none

Which of the following is the most likely explanation for this patient's oliguria?

- ☐ A) Decreased intravascular oncotic pressure
- ☐ B) Decreased intravascular volume
- ☐ C) Deposition of myoglobin in distal tubules
- ☐ D) Disseminated intravascular coagulation
- ☐ E) Toxin-mediated endothelial injury



40. A 12-year-old girl with a 5-year history of type 1 diabetes mellitus is brought for a follow-up examination. Over the past 4 weeks, she has had persistent sweating, shakiness, and pallor despite a 60% decrease in her insulin doses. Her blood glucose concentrations continue to range between 40 and 59 mg/dL throughout the day. Her exercise regimen and dietary intake have not changed. Examination shows no abnormalities. Serum studies show:

Na ⁺	137 mEq/L
Cl ⁻	102 mEq/L
K ⁺	4.2 mEq/L
HCO ₃ ⁻	24 mEq/L
Glucose	45 mg/dL
Insulin	increased
C peptide	not detectable
Glucagon	increased
Cortisol	increased

Which of the following is the most likely cause of these findings?

- ☐ A) Decreased growth hormone production
- ☐ B) Excess exogenous insulin
- ☐ C) Increased endogenous insulin production
- ☐ D) Increased epinephrine production
- ☐ E) Insulin resistance

41. A previously healthy 10-year-old girl is brought to the emergency department 2 hours after the onset of pain in her right knee. She also has had a decreased appetite for 12 hours. There is no history of trauma to the knee or hip. Her temperature is 38.9°C (102°F). Examination shows erythema, swelling, and warmth of the right knee. Passive range of motion of the knee is limited by pain; the associated muscles are in spasm. She walks with an obvious limp. Which of the following is the most likely diagnosis?
- ☐ A) Juvenile idiopathic arthritis
 - ☐ B) Osteomyelitis
 - ☐ C) Septic arthritis
 - ☐ D) Transient synovitis of the hip
 - ☐ E) Transient synovitis of the knee



42. An 18-year-old woman is brought to the emergency department 1 hour after she attempted suicide by ingesting a hydrochloric acid solution. She has substernal chest pain. Her temperature is 37.6°C (99.7°F), pulse is 110/min, respirations are 20/min, and blood pressure is 135/85 mm Hg. The lungs are clear to auscultation. The abdomen is soft, nondistended, and nontender; bowel sounds are normal. Which of the following is the most appropriate next step in management?
- ☐ A) Administration of sodium bicarbonate solution
 - ☐ B) Barium swallow
 - ☐ C) CT scan of the chest
 - ☐ D) Esophagoscopy
 - ☐ E) Placement of a nasogastric tube



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43. A 5-year-old boy is brought to the emergency department because of a 1-day history of fever, nausea, vomiting, pain during urination, and a widespread, red, bumpy rash. Three days ago, he began amoxicillin therapy for exudative pharyngitis. He appears ill. His temperature is 39°C (102.2°F), pulse is 120/min, respirations are 20/min, and blood pressure is 95/48 mm Hg. A photograph of the rash is shown. There is anterior cervical lymphadenopathy bilaterally. Examination shows bilateral conjunctival erythema and photophobia. There are several 3- to 5-mm oral ulcers. Which of the following is the most likely diagnosis?
- ☐ A) Mucocutaneous lymph node syndrome (Kawasaki disease)
 - ☐ B) Scarlet fever



43. A 5-year-old boy is brought to the emergency department because of a 1-day history of fever, nausea, vomiting, pain during urination, and a widespread, red, bumpy rash. Three days ago, he began amoxicillin therapy for exudative pharyngitis. He appears ill. His temperature is 39°C (102.2°F), pulse is 120/min, respirations are 20/min, and blood pressure is 95/48 mm Hg. A photograph of the rash is shown. There is anterior cervical lymphadenopathy bilaterally. Examination shows bilateral conjunctival erythema and photophobia. There are several 3- to 5-mm oral ulcers. Which of the following is the most likely diagnosis?
- ☐ A) Mucocutaneous lymph node syndrome (Kawasaki disease)
 - ☐ B) Scarlet fever
 - ☐ C) Serum sickness
 - ☐ D) Staphylococcal scalded-skin syndrome
 - ☐ E) Stevens-Johnson syndrome

44. An 11-month-old boy is brought to the emergency department because of a 2-day history of fever, cough, and rash. He was born at term after an uncomplicated pregnancy and delivery. He has had frequent episodes of otitis media since birth and was hospitalized at the age of 6 months for meningitis. He is at the 10th percentile for length and 5th percentile for weight. His temperature is 39°C (102.2°F) , pulse is 140/min, respirations are 40/min, and blood pressure is 86/56 mm Hg . Examination shows an erythematous scaly rash with several weeping lesions over the trunk and extremities. There is bilateral scarring of the tympanic membranes and dried blood and crusty material in the nares. Decreased breath sounds are heard on the right. Laboratory studies show:

Hemoglobin	10.3 g/dL
Leukocyte count	12,400/mm ³
Platelet count	75,000/mm ³

An x-ray of the chest shows a right middle lobe consolidation. Which of the following is the most likely diagnosis?

- ☐ A) Chronic granulomatous disease
- ☐ B) HIV infection
- ☐ C) Severe combined immunodeficiency
- ☐ D) Thymic-parathyroid dysplasia (DiGeorge syndrome)
- ☐ E) Wiskott-Aldrich syndrome

45. A 16-year-old boy is brought to the physician for a follow-up examination. Six months ago, he noticed a lump in his right testicle after participating in a soccer game, during which he sustained a kick to the groin. Ultrasonography 2 days later showed a cystic lesion consistent with trauma. He was advised to avoid contact sports for 2 weeks and was told that the lesion would resolve without treatment. The patient reports that since that time, the lump has increased in size and has become firm. He also reports facial acne, swelling of his breasts, stretch marks over his abdomen, and a 4.5-kg (10-lb) weight gain despite eating no more than usual. He has not had respiratory, gastrointestinal, or urinary symptoms. He appears well. Vital signs are within normal limits. Examination shows facial acne. There are striae over the lower abdomen. A firm, fixed, 6-cm, nontender mass is palpated in the right testicle. There is no inguinal lymphadenopathy. Measurement of which of the following is the most appropriate next step in diagnosis?

- ☐ A) Serum alkaline phosphatase activity
- ☐ B) Serum CA 125 concentration
- ☐ C) Serum α -fetoprotein and β -hCG concentrations
- ☐ D) Serum prostate-specific antigen concentration
- ☐ E) Urine homovanillic–vanillylmandelic acid

46. The mother of a 12-hour-old newborn expresses concern about the shape of his head. He was born at term after an uncomplicated pregnancy. The second stage of labor was 18 hours and low forceps were used to complete delivery. Apgar scores were 8 and 9 at 1 and 5 minutes, respectively. His temperature is 36.8°C (98.2°F), pulse is 128/min, and respirations are 32/min. He appears vigorous and alert. Examination shows prominent molding of the head. There is a boggy 5 x 7-cm area of swelling that extends across the sagittal suture from the right parieto-occipital area to the left parietal area. Which of the following is the most likely diagnosis?
- ☐ A) Caput succedaneum
 - ☐ B) Cephalhematoma
 - ☐ C) Epidural hematoma
 - ☐ D) Intracerebral hematoma
 - ☐ E) Subdural hematoma

47. A 3-year-old boy is brought to the physician by his parents because of a bowleg deformity that has become more noticeable during the past year. He was adopted at the age of 8 months. The birth parents were told that he had hypotonia as a newborn, but no other medical history was provided. He sat upright at the age of 12 months and walked at the age of 24 months. He is at the 3rd percentile for height and 20th percentile for weight. Examination shows frontal bossing and a small nasal bridge, a normal trunk, short limbs, and increased lumbar lordosis. There is genu varum of the lower extremities. Active range of motion of the knees is full, and there is no bony tenderness. X-rays of the lower extremities show metaphyseal flaring, inverted V-shape distal femoral physis, and a disproportionately long fibula. The most likely explanation for these findings is a defect of which of the following?
- ☐ A) Cartilage oligomeric matrix protein
 - ☐ B) Fibroblast growth factor 3
 - ☐ C) Sulfate transport protein
 - ☐ D) Type 1 collagen
 - ☐ E) Type 2 collagen

48. A 7-year-old boy is brought to the office in the fall for a routine examination. He has mild intermittent asthma and gastroesophageal reflux disease. His medications are albuterol and ranitidine. Immunizations were up-to-date when he entered kindergarten. He is at the 75th percentile for height and 50th percentile for weight. Vital signs are within normal limits. Examination shows no abnormalities. Which of the following is the most appropriate vaccine to administer?
- ☐ A) Diphtheria-tetanus-acellular pertussis (DTaP)
 - ☐ B) *Haemophilus influenzae* type b
 - ☐ C) Influenza
 - ☐ D) Meningococcal
 - ☐ E) Pneumococcal



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49. A 15-year-old girl is brought to the physician because of a 1-week history of constant severe abdominal pain. During the past year, she has had 10 episodes of cramps; each episode lasts 3 to 5 days. She has no history of serious illness and takes no medications. She has never had a menstrual period. She is sexually active with one male partner and uses condoms inconsistently. She is at the 80th percentile for height and weight. Her temperature is 37.8°C (100°F), pulse is 120/min, respirations are 16/min, and blood pressure is 90/50 mm Hg. Breast and genital development is sexual maturity rating stage 5. Abdominal examination shows moderate tenderness. A mass is palpated in the suprapubic region at the midline. Pelvic examination shows normal-appearing external genitalia and lower vagina. The cervix cannot be visualized because of bluish bulging vaginal tissue that obscures the upper vagina. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Karyotype analysis
- ☐ B) Urine culture
- ☐ C) Intravenous pyelography
- ☐ D) Pelvic ultrasonography
- ☐ E) Laparoscopy

50. A 14-month-old girl is brought to the physician because of intermittent vomiting for 12 hours. During this period, her playing and sleeping have been interrupted by episodes of crying and fretfulness that last 20–30 minutes. Over the past hour she has had two bowel movements, one containing blood. Examination shows no abdominal distention. A noncontrast x-ray of the abdomen shows no evidence of bowel obstruction. Which of the following is the most likely diagnosis?

- ☐ A) Appendicitis
- ☐ B) Dysentery
- ☐ C) Intussusception
- ☐ D) Viral gastroenteritis
- ☐ E) Volvulus



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